

Few-Trial Damage Recovery with Quality-Diversity Repertoires

Repertoires built with and without reinforcement learning, adapted with Intelligent Trial & Error

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<https://github.com/AnaisAzouaoui/qd-damage-recovery>

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Abstract

A legged robot that loses the use of a leg should find a new gait with as few physical trials as possible. Intelligent Trial & Error (ITE) does so by searching, with Bayesian optimisation, a repertoire of diverse gaits learned beforehand in simulation. Recent Quality-Diversity (QD) algorithms build such repertoires with the help of reinforcement-learning (RL) critics and reach much higher performance, but it is not known whether they keep the diversity that makes a repertoire useful after damage. On the Brax Ant, we build repertoires at equal budget with MAP-Elites, PGA-ME and DCRL-ME (3 seeds each), measure their resilience to 12 damage scenarios, and adapt them with ITE, simple baselines and RL re-training (TD3), counting episodes on the damaged robot. Repertoires built with RL are as resilient as those of MAP-Elites in relative terms and keep much better gaits in absolute terms. ITE needs a median of 5 trials, recovers 33% of the progress of the best intact gait (against 18% without adaptation), but is barely better than trying gaits in decreasing order of their intact performance: its small advantage after three trials is gone after five. Except in one case where the intact policy already matched ITE before any training, TD3 needs more than a thousand episodes to reach the same performance, when it reaches it at all.

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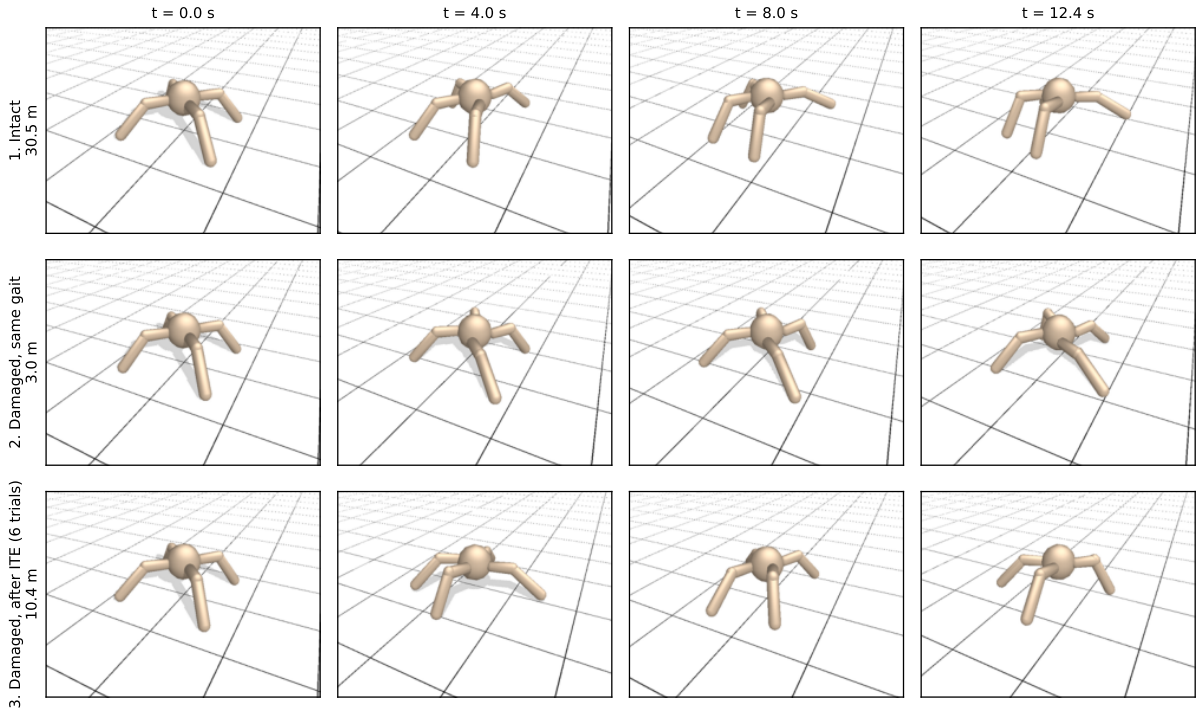


Figure 1: The simulated quadruped with its best gait (top), the same gait after the front-right leg is paralysed (middle), and the gait found by ITE in a few trials (bottom). Distances travelled in 12.5 s on the left.

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1 Introduction

Animals keep walking after an injury because they already know many ways to move: they limp, shift their weight, use another leg more. Cully et al. [5] gave robots the same ability. *Before* any damage, a large repertoire of diverse gaits is learned in simulation with MAP-Elites [9]; *after* damage, the robot runs Intelligent Trial & Error (ITE): it tests a promising gait, observes how well it works, and updates a Gaussian-process model of the performance of *all similar* gaits, so that each trial is informative about many of them. A damaged hexapod recovered in less than two minutes.

Since then, QD algorithms have been combined with deep RL. PGA-ME [10] improves half of the offspring with the gradient of a TD3 critic [7]; DCRL-ME [6] conditions critic and actor on the targeted behaviour descriptor, so that RL can improve a gait *without changing its style*.

These methods produce much better gaits. Whether they also produce repertoires that are useful after damage is less clear: RL pushes towards high performance, possibly at the cost of the diversity that adaptation needs. Moreover, adaptation has mostly been evaluated with an *oracle* that tries every skill on the damaged robot [2], which says nothing about the number of trials a real robot would need.

Summary of results (i) At equal budget, DCRL-ME covers the most cells (83.8%) and PGA-ME finds the fastest gaits (best fitness 1922). The *relative* resilience to damage does not differ significantly between algorithms, but the best gait left after damage is much better in repertoires built with RL. (ii) ITE stops after a median of 5 trials and recovers 33% of the intact progress, against 18% for the best intact gait; however, a model-free strategy that tries gaits in decreasing order of intact performance (top- k) does almost as well: ITE is slightly ahead after three trials, mostly on DCRL-ME repertoires, and no longer after five. (iii) RL re-training needs more than a thousand episodes, and starting from the intact policy does not help. (iv) Searching the continuous descriptor space with the DCRL-ME conditioned actor, instead of the repertoire cells, is clearly worse.

Questions

- Q1.** At equal budget, are repertoires built with RL (PGA-ME, DCRL-ME) more or less *resilient* to damage than those of MAP-Elites?
- Q2.** How many *trials* does ITE need on each kind of repertoire, compared with simpler strategies?
- Q3.** How does this compare with re-training a policy by RL on the damaged robot, counted in episodes?

2 Background and related work

2.1 Quality-Diversity and MAP-Elites

A QD algorithm searches for a *collection* of solutions that are both diverse and high-performing. MAP-Elites [9] discretises a user-defined *behaviour descriptor* space into cells and keeps, in each cell, the best solution found (its *elite*). At each iteration, elites are selected at random, mutated, evaluated, and inserted in their cell if they beat its current occupant.

2.2 QD with reinforcement learning

Mutations are blind. PGA-ME [10] trains a TD3 critic on all transitions collected by the population and uses its gradient to improve half of the offspring. DCG-ME and DCRL-ME [6] condition the critic on a target descriptor, so that the gradient improves a policy while keeping its behaviour, and also train a descriptor-conditioned actor whose outputs are added to the repertoire. Chalumeau et al. [2] compared QD methods with skill-discovery RL (DIAYN, DADS, SMERL) on few-shot adaptation tasks, including a leg dysfunction on Ant: after perturbation, every skill is evaluated and the best one is kept. QD methods were more robust to large perturbations, and PGA-ME adapted better than MAP-Elites.

2.3 Damage recovery with repertoires

ITE [5] uses the M-BOA algorithm described in Section 4.2. Follow-ups removed the need for resets [4], selected among several priors [8], or used hierarchical repertoires [1]. We keep the original algorithm and study the effect of *how the repertoire was built*.

3 Experimental setup

3.1 Robot, task and behaviour descriptor

We use the Brax Ant (`ant_uni` in QDax [3]): a torso and four legs, each with a hip and an ankle actuator (8 actions). The reward is the forward velocity plus a survival bonus, minus control and contact costs, over episodes of 250 steps (12.5 s). As in the DCRL-ME notebook, rewards are shifted by a constant and clipped at zero for all algorithms. The descriptor is the fraction of time each foot touches the ground, $\mathbf{d} \in [0, 1]^4$; the repertoire is a centroidal Voronoi tessellation of 1024 cells.

Leg mapping The mapping from actions to legs was checked in three independent ways (geometry of the bodies at reset, relative motion of each foot when a single action is applied, and video): actions (0, 1), (2, 3), (4, 5), (6, 7) drive the hip and ankle of the front-left (FL), hind-left (HL), hind-right (HR) and front-right (FR) legs.

3.2 Algorithms and budget

The three algorithms share the environment, the policy architecture (MLP, two hidden layers of 128 units, tanh output), the repertoire, and the number of evaluations per iteration (256); hyper-parameters specific to each algorithm are those of the official QDax notebooks (Table 1). Each run lasts 2000 iterations, i.e. 512,256 episodes; 3 seeds per algorithm. A test checks that the three configurations only differ by their emitter.

Table 1: Main hyper-parameters (full configuration in `configs/`).

	MAP-Elites	PGA-ME	DCRL-ME
Offspring per iteration	256 mutations	128 mutations + 128 PG	128 + 64 PG + 64 actor
Iso/line variation σ	0.005 / 0.05	0.005 / 0.05	0.005 / 0.05
Critic	–	TD3, 256×256	descriptor-cond., 256×256
Critic steps / iteration	–	300	3000
PG steps / iteration	–	100	150

3.3 Damage scenarios

A damage multiplies the actions of some joints by a factor λ ($\lambda = 0$: paralysed, $\lambda = 0.5$: weakened). Since the torque of an Ant actuator is proportional to its action, this is equivalent to scaling its strength, as in Chalumeau et al. [2]. We use 12 scenarios: each leg paralysed, each leg weakened, two legs paralysed (FL+HL, same side; FL+FR, both front legs), and one ankle paralysed (FL, HR).

3.4 Performance scale

Episodes are noisy (random initial state), and a robot that does not move still collects the survival bonus: its return F_0 was measured at 4.23 per step (1058.7 per episode). We report

$$p = \frac{F - F_0}{S}, \quad S = 1.2 \text{ per step (300 per episode)},$$

so that $p \approx 0$ for a standing robot and $p \approx 1$ for ≈ 15 m in 12.5 s, or percentages of the progress $F - F_0$ of the best intact gait. With this scale, the variance of a single episode’s performance is $\approx 10^{-3}$, the value used by Cully et al. [5].

4 Methods

4.1 Oracle and resilience

For each repertoire and damage, every policy is evaluated on 4 episodes. The best mean is the *oracle* performance, an upper bound for any method that picks a gait from the repertoire (it needs 4 episodes for each of the 731–862 policies of a repertoire, for every damage). The *resilience* of a repertoire to a damage is the fraction of the intact oracle progress still reachable: $(F_{\text{damaged}}^* - F_0)/(F_{\text{intact}}^* - F_0)$.

4.2 Intelligent Trial & Error

Let $P(\mathbf{d}_i)$ be the prior performance of cell i on the intact robot. ITE models the residual between observed and prior performance with a Gaussian process with a Matérn-5/2 kernel on descriptors:

$$\begin{aligned}\mu_t(\mathbf{d}) &= P(\mathbf{d}) + \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1} (\mathbf{y} - P(\mathbf{D}_t)), \\ \sigma_t^2(\mathbf{d}) &= \sigma_f^2 - \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1} \mathbf{k}(\mathbf{d}),\end{aligned}$$

where $\mathbf{D}_t$ and $\mathbf{y}$ are the descriptors and observed performances of the gaits tried so far. The next gait maximises the upper confidence bound $\mu_t + \kappa\sigma_t$; ITE stops when the best observed performance reaches $\alpha \max_{\mathbf{d}} \mu_t(\mathbf{d})$. One trial is *one* episode on the damaged robot.

Prior The fitness stored in a repertoire comes from a single noisy episode, and MAP-Elites keeps the luckiest evaluations: elites are over-estimated (on development repertoires, the correlation between stored and re-evaluated fitness is 0.88, and the “best” cell of a MAP-Elites repertoire performs worse than a standing robot once re-evaluated). As the repertoire is built offline in simulation, we re-evaluate it on the intact robot (4 episodes) and use this as prior; the best-intact baseline uses the same values.

Hyper-parameters With the values of Cully et al. [5] (length-scale 0.4, unit signal variance, noise 10^{-3}), ITE stopped after very few trials and plateaued. The residuals have a variance of 0.03–0.06, not 1, and a large nugget (neighbouring cells differ). We therefore fitted $(\ell, \sigma_f^2, \sigma_n^2)$ by maximising the marginal likelihood on *development* repertoires (short runs of the three algorithms, never used for the results): $\ell = 0.4$, $\sigma_f^2 = 0.05$, $\sigma_n^2 = 0.02$ (log-likelihood +4059 against −1280 for the original values). $\kappa = 0.05$ and $\alpha = 0.9$ are those of Cully et al. [5]; values of κ from 0.05 to 2 made little difference on the development set.

4.3 Baselines

Best-intact: replay the best intact gait (no adaptation, one trial). *Top-k*: try cells by decreasing prior, without a model. *Random*: try cells at random. All methods run for 20 trials and are repeated 5 times per (repertoire, damage); the quality of the recommended gait (best observed so far) is measured with its oracle mean.

4.4 Statistics

The repetitions of an adaptation only differ by episode noise, so they are averaged first: the statistical unit is a (repertoire, damage) pair, i.e. 36 units per algorithm. Algorithms are compared with two-sided Mann-Whitney tests on these units, and two adaptation methods with paired two-sided Wilcoxon signed-rank tests. Units of the same seed share a repertoire and are not fully independent, so p-values should be read as indicative. Medians in the tables are taken over all repetitions.

4.5 RL re-training

TD3 (QDax implementation) is trained on the damaged robot, either from scratch or starting from the best intact DCRL-ME policy (same architecture; the critic starts from scratch), on 4 damages (FR leg paralysed, HL leg weakened, FL+FR legs paralysed, FL ankle paralysed) and 2 seeds, for 2000 episodes with 16 robots in parallel and one gradient step per transition. The deterministic policy is evaluated on 8 episodes after every 16 training episodes. In the reported runs the starting policy is the elite with the highest *stored* fitness, because the oracle ran in parallel; for seed 1 it is also the best re-evaluated elite, for seed 0 it is a different elite whose re-evaluated intact fitness is 1.4% below the best one. The code now starts from the best re-evaluated elite.

5 Results

5.1 Repertoires on the intact robot

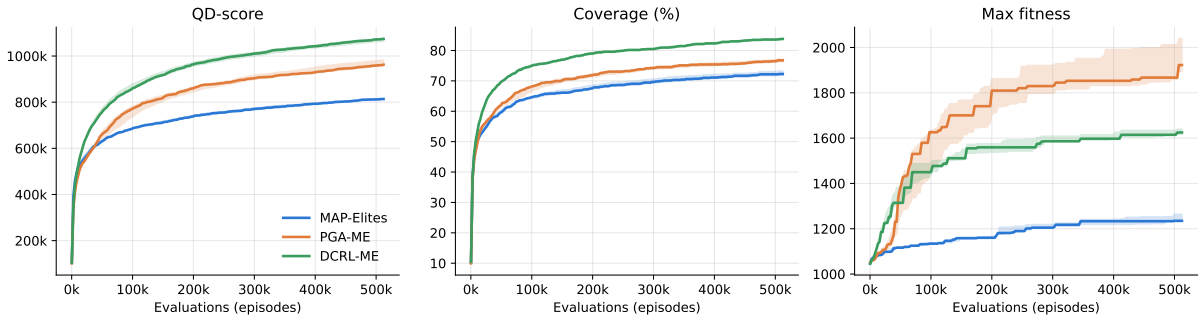


Figure 2: QD-score, coverage and best fitness during training (median and inter-quartile range over 3 seeds).

Table 2: Repertoires at the end of training (median over seeds).

Algorithm	Coverage (%)	Max fitness	QD-score ($\times 10^3$)
MAP-Elites	72.3	1235	814
PGA-ME	76.8	1922	962
DCRL-ME	83.8	1625	1074

Both RL-based algorithms improve over MAP-Elites at equal budget (Figure 2 and Table 2), in different ways. DCRL-ME fills the most cells and reaches the highest QD-score, which is consistent with its descriptor-conditioned critic improving gaits *without changing their style*. PGA-ME finds by far the fastest gaits (median best fitness 1922 against 1625 for DCRL-ME and 1235 for MAP-Elites): its unconditioned critic pushes offspring towards high reward, wherever they land in the grid. The fastest gait we inspected (PGA-ME) is a bounding gait in which the feet touch the ground only briefly. The stored fitness of these elites is optimistic (Section 6).

5.2 Resilience to damage (Q1)

In relative terms, the three kinds of repertoires are equally resilient: the median share of the intact progress that remains reachable after damage is 54% for MAP-Elites, 73% for PGA-ME and 64% for DCRL-ME, and none of the pairwise differences is significant (Mann-Whitney, $p = 0.532$, $p = 0.996$ and $p = 0.496$; Figure 3). We therefore find no sign that RL makes repertoires less useful after damage. In absolute terms, the difference is large (Table 3): after damage, the best

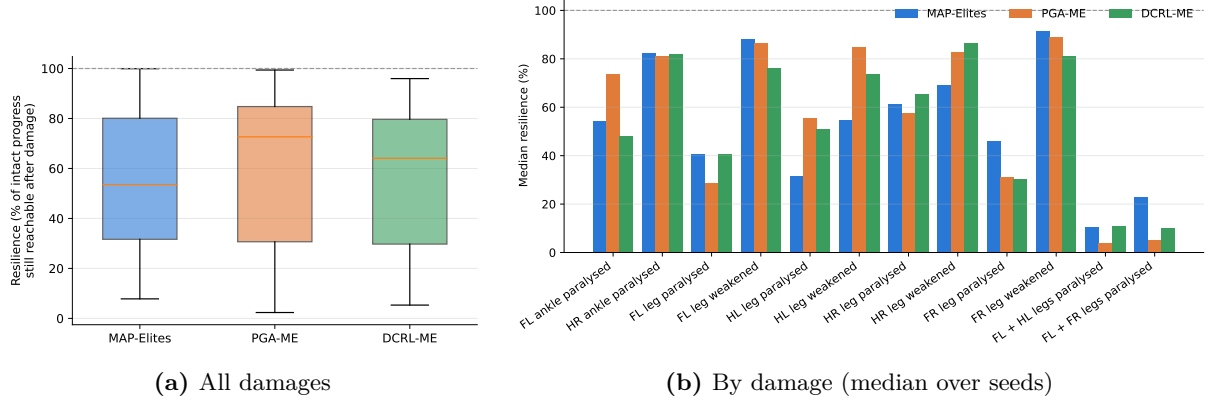


Figure 3: Resilience: share of the intact oracle progress still reachable after damage.

Table 3: Resilience (%) and absolute oracle performance p after damage (median over seeds).

Damage	MAP-Elites (%)	PGA-ME (%)	DCRL-ME (%)	MAP-Elites (p)	PGA-ME (p)	DCRL-ME (p)
FL ankle paralysed	54	74	48	0.28	1.86	0.83
HR ankle paralysed	82	81	82	0.35	2.06	1.48
FL leg paralysed	41	29	41	0.21	0.73	0.73
FL leg weakened	88	87	76	0.38	2.20	1.32
HL leg paralysed	32	55	51	0.13	1.72	0.92
HL leg weakened	55	85	74	0.24	2.33	1.32
HR leg paralysed	61	58	65	0.23	1.79	1.18
HR leg weakened	69	83	87	0.28	2.15	1.56
FR leg paralysed	46	31	30	0.20	0.97	0.54
FR leg weakened	91	89	81	0.40	2.25	1.42
FL + HL legs paralysed	11	4	11	0.06	0.10	0.19
FL + FR legs paralysed	23	5	10	0.12	0.16	0.18

gait available is worth a median $p = 1.84$ in PGA-ME repertoires, $p = 1.14$ in DCRL-ME ones and only $p = 0.23$ in MAP-Elites ones. Weakening a leg is usually mild; paralysing a whole leg removes roughly half of the progress; with two paralysed legs, no repertoire keeps a useful gait.

5.3 Adaptation in a few trials (Q2)

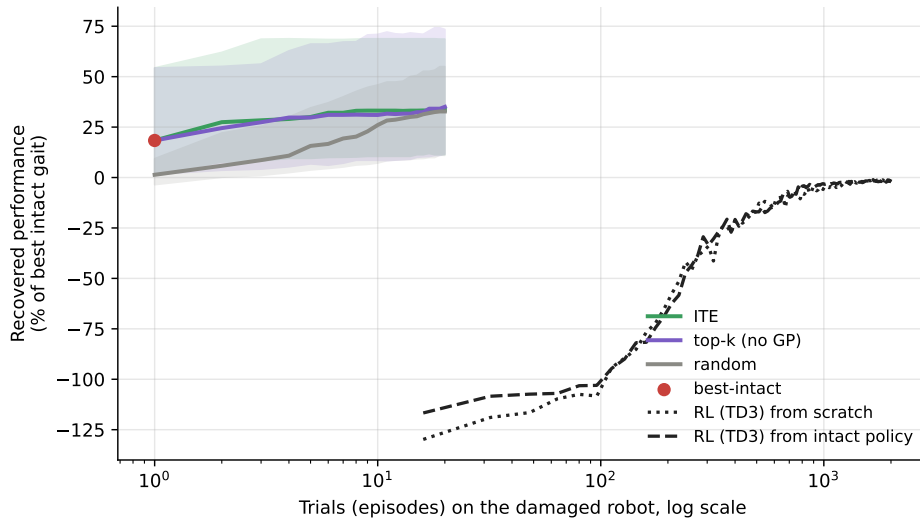


Figure 4: Performance of the recommended gait against the number of trials on the damaged robot (median and IQR over repertoires, damages and repetitions; each repertoire relative to its own best intact gait). RL curves are relative to the best intact DCRL-ME gait.

Table 4: Adaptation methods (median over repertoires, damages and repetitions). “% intact”: share of the progress of the best intact gait; “% oracle”: share of the best gait available after damage.

Method	Trials	% intact	% oracle	Absolute p
ITE	5	33	62	0.37
top- k (no GP)	20	35	77	0.40
random	20	33	60	0.31
best-intact	1	18	34	0.19

Table 5: Share of the intact progress (%) recovered after a given number of trials (median).

Method	1 trial	3 trials	5 trials	10 trials	20 trials
ITE	18	28	30	33	33
top- k (no GP)	18	27	30	31	35
random	1	9	16	26	33

ITE meets its stopping criterion in 89% of the runs, after a median of 5 trials (IQR 2–10). The recommended gait recovers 33% of the progress of the best intact gait, against 18% when the best intact gait is simply replayed, and 62% of what an exhaustive search of the repertoire would find (Table 4). Random search is much slower (9% after three trials, against 28% for ITE).

The comparison with top- k , which tries gaits in decreasing order of intact performance without any model, is less favourable: after three trials ITE reaches 28% and top- k 27%; after 20 trials, 33% and 35% (Figure 4 and Table 5). Paired over the 108 (repertoire, damage) units, ITE is better than top- k after five trials in 47% of them, equal in 5% and worse in 48% (Wilcoxon $p = 0.445$). After three trials ITE has a small advantage (mean difference +0.05 in p , $p = 0.009$), which comes mostly from DCRL-ME repertoires (+0.13, $p = 0.019$) and disappears after five and ten trials (Table 6). The pooled medians of Table 7 suggest a larger gap on DCRL-ME repertoires (0.59 against 0.48 after ten trials), but the paired test does not support it ($p = 0.407$): unit by unit, ITE is better as often as it is worse (Table 6). On PGA-ME repertoires, the best intact gait already keeps most of its performance ($p = 1.43$ at the first trial). On MAP-Elites repertoires, all methods stay close to a standing robot.

Table 6: ITE against top- k at equal numbers of trials: mean difference in absolute performance p (ITE – top- k), number of units where ITE is better / equal / worse, and paired Wilcoxon test, over the (repertoire, damage) units.

Repertoire	3 trials	5 trials	10 trials
MAP-Elites	+0.01 (18/0/18; $p = 0.581$)	+0.00 (16/0/20; $p = 0.798$)	+0.01 (20/0/16; $p = 0.451$)
PGA-ME	+0.03 (19/4/13; $p = 0.135$)	−0.03 (18/1/17; $p = 0.974$)	−0.03 (18/0/18; $p = 0.944$)
DCRL-ME	+0.13 (20/1/15; $p = 0.019$)	+0.08 (17/4/15; $p = 0.166$)	+0.04 (17/2/17; $p = 0.407$)

What a robot finally gets depends mostly on how its repertoire was built (Table 8): with ITE, the recommended gait is worth $p = 1.47$ with PGA-ME, $p = 0.59$ with DCRL-ME and $p = 0.11$ with MAP-Elites (Mann-Whitney on the (repertoire, damage) units: $p < 0.001$ for MAP-Elites against PGA-ME, $p < 0.001$ against DCRL-ME and $p = 0.002$ between PGA-ME and DCRL-ME; at the level of seeds, every PGA-ME repertoire is above every DCRL-ME repertoire, itself above every MAP-Elites repertoire). Figures 5 and 6 show one ITE run.

Table 7: Absolute performance p of the recommended gait at equal numbers of trials, by repertoire (median).

Repertoire – method	1 trial	3 trials	5 trials	10 trials	20 trials
MAP-Elites – ITE	0.07	0.10	0.11	0.11	0.12
MAP-Elites – top- k	0.07	0.09	0.10	0.09	0.11
MAP-Elites – random	-0.03	0.02	0.07	0.09	0.12
PGA-ME – ITE	1.43	1.47	1.47	1.47	1.47
PGA-ME – top- k	1.43	1.47	1.52	1.52	1.57
PGA-ME – random	0.10	0.21	0.47	0.78	1.11
DCRL-ME – ITE	0.22	0.47	0.47	0.59	0.60
DCRL-ME – top- k	0.22	0.36	0.37	0.48	0.59
DCRL-ME – random	0.03	0.22	0.28	0.49	0.59

Table 8: ITE on each kind of repertoire (median over seeds, damages and repetitions).

Repertoire	Trials	% intact	Absolute p
MAP-Elites	6	24	0.11
PGA-ME	3	55	1.47
DCRL-ME	6	33	0.59

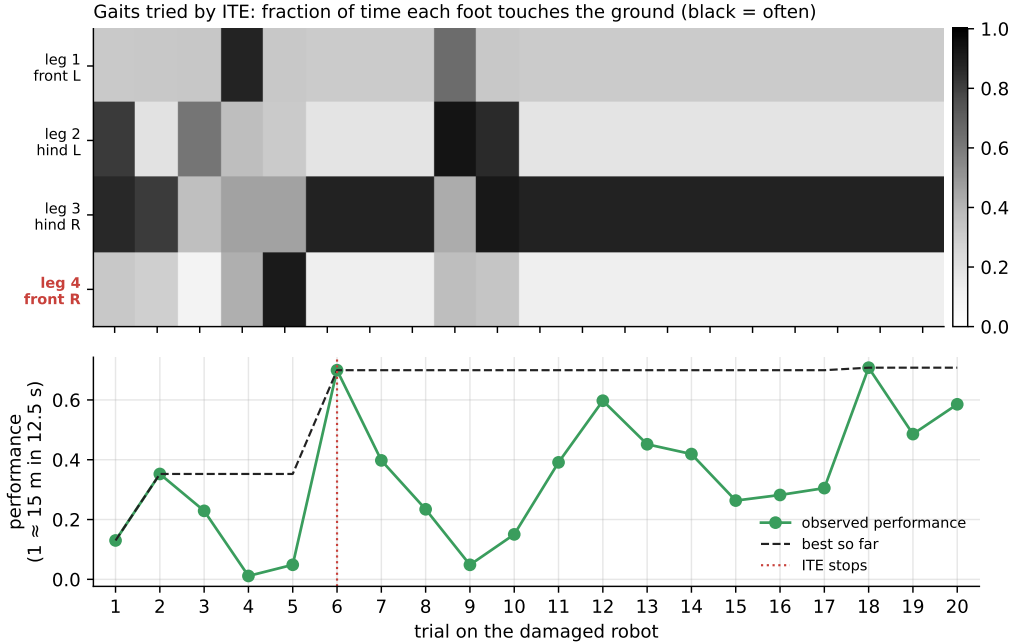


Figure 5: One ITE run on a DCRL-ME repertoire (front-right leg paralysed; the five repetitions all recommend the same gait, after 6 or 7 trials, and this one used 6): feet-contact profile of each gait tried, and its observed performance.

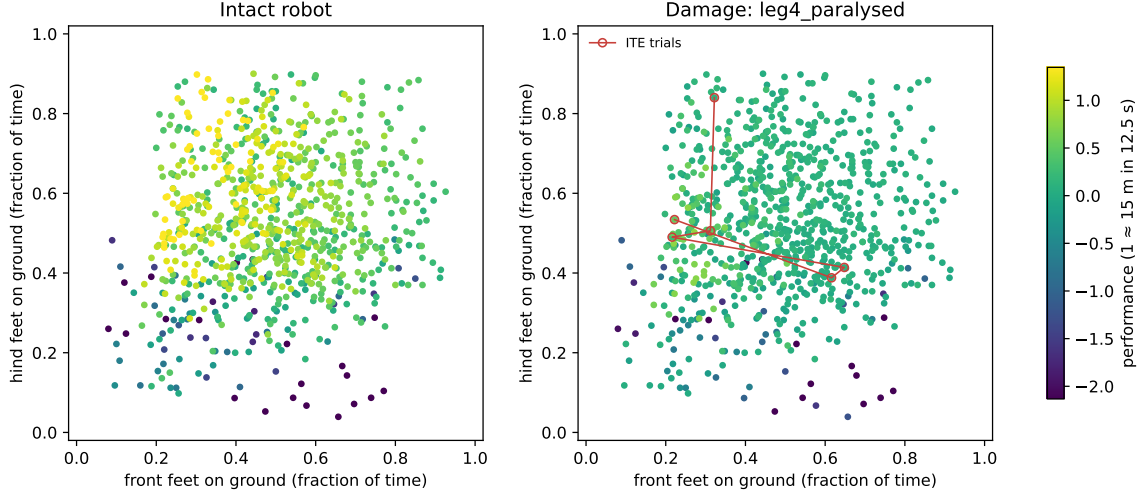


Figure 6: The same repertoire before and after damage (front vs hind feet contact), and the gaits tried by ITE.

5.4 Comparison with RL re-training (Q3)

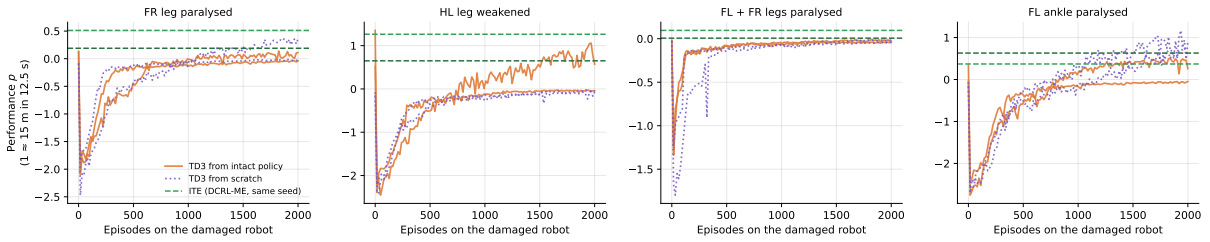


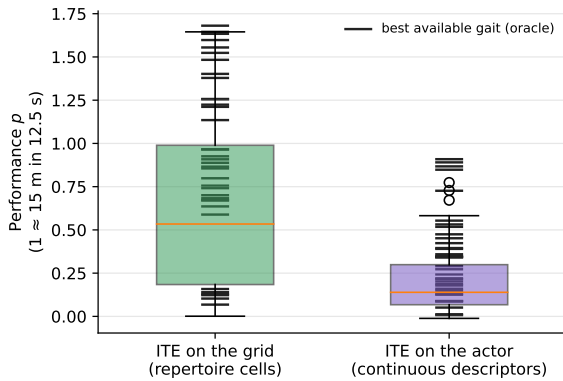
Figure 7: TD3 on the damaged robot, from scratch and from the best intact DCRL-ME policy (two seeds each), against the performance of the gait found by ITE on the DCRL-ME repertoire of the same seed (dashed lines).

Within 2000 episodes, TD3 from scratch matches the gait found by ITE in 3/8 runs (median 1264 episodes when it does), and TD3 from the intact policy in 3/8 runs (Figure 7 and Table 9), against 1 to 20 trials for ITE. In one of the latter cases, the intact policy already matched ITE before any training. Otherwise, starting from the intact policy does not help: its performance collapses at the first evaluation (after 16 episodes), and the robot has to learn again almost from scratch. The most likely cause is that the policy starts following the gradient of a new critic that has not been trained yet; we did not test it. A pre-trained critic, a warm-up period without policy updates, or a smaller policy learning rate could make fine-tuning useful.

Table 9: Episodes needed by TD3 to match the gait found by ITE (DCRL-ME repertoire of the same seed); “–”: not reached within 2000 episodes.

Damage	RL	Seed	ITE trials	RL episodes to match	RL budget
FL ankle paralysed	from intact policy	0	10	1264	2000
FL ankle paralysed	from intact policy	1	6	–	2000
FL ankle paralysed	from scratch	0	10	1072	2000
FL ankle paralysed	from scratch	1	6	1808	2000
HL leg weakened	from intact policy	0	1	0	2000
HL leg weakened	from intact policy	1	6	1536	2000
HL leg weakened	from scratch	0	1	–	2000
HL leg weakened	from scratch	1	6	–	2000
FR leg paralysed	from intact policy	0	7	–	2000
FR leg paralysed	from intact policy	1	15	–	2000
FR leg paralysed	from scratch	0	7	–	2000
FR leg paralysed	from scratch	1	15	1264	2000
FL + FR legs paralysed	from intact policy	0	19	–	2000
FL + FR legs paralysed	from intact policy	1	20	–	2000
FL + FR legs paralysed	from scratch	0	19	–	2000
FL + FR legs paralysed	from scratch	1	20	–	2000

5.5 Bonus: searching the continuous descriptor space



ITE search space	Trials	Found p	Oracle p
Grid (repertoire cells)	6	0.53	0.90
Continuous (conditioned actor)	3	0.14	0.32

Figure 8: ITE on the cells of a DCRL-ME repertoire against ITE on 2048 gaits generated by its descriptor-conditioned actor for targets spread over $[0, 1]^4$ (3 dedicated runs, same runs for both).

DCRL-ME also learns an actor $\pi(a \mid o, \mathbf{d})$ that should produce a gait with any requested feet-contact profile $\mathbf{d}$. We let ITE search 2048 such gaits instead of the repertoire cells, on 3 dedicated DCRL-ME runs where the actor was saved. The answer is clearly negative (Figure 8): the gaits produced by the actor are much weaker than the elites (best available gait after damage $p = 0.32$ against $p = 0.90$, and never better in any of the (run, damage) pairs), and ITE on the actor finds $p = 0.14$ against $p = 0.53$ on the grid (better in only 3 of the 36 (run, damage) pairs; Wilcoxon $p < 0.001$). The actor is a useful source of offspring during training, but it does not replace the elites that were selected and refined.

6 Discussion

Why ITE is not much better than top- k ITE helps when trying one gait tells something about *similar* gaits. Here, the effect of damage is only weakly correlated between neighbouring cells of the feet-contact descriptor: on development repertoires, a large part of the variance of the residuals is already present between the closest cells (the “nugget” of the variogram, from a few

percent of the variance for MAP-Elites to more than half for PGA-ME), and the noise variance fitted by maximum likelihood (0.02) is not small compared with the signal variance (0.05). The Gaussian process can therefore generalise little, while the re-evaluated intact performance is already a good guide. A descriptor closer to what a damage affects (e.g. the torque or contact pattern of each leg separately) might make the model more useful.

Noisy elites A repertoire stores the fitness of a single noisy episode, and MAP-Elites keeps the luckiest ones: elites are over-estimated. Re-evaluating the repertoire before damage, offline in simulation, is cheap and was necessary for both ITE and the best-intact baseline. This is the problem studied by uncertain QD.

The oracle is optimistic The oracle is the maximum of noisy means over ~ 1000 cells, so it is biased upwards; percentages of the oracle under-estimate all methods.

Counting episodes, not compute In simulation, RL consumes thousands of episodes in minutes; on a real robot, each episode costs time, wear and supervision. We therefore compare trials and episodes, not wall-clock time.

7 Limitations

Simulation only; one robot; damages modelled as action scaling; one repertoire size and one budget; 3 seeds per algorithm, so the (repertoire, damage) units used in the tests are not independent and the p-values are indicative; ITE hyper-parameters fitted on development repertoires of the same algorithms; RL re-training with standard TD3 settings only (no tuning of the fine-tuning variant), 2 seeds and 4 damages.

8 Conclusion and future work

Repertoires built with the help of RL critics are not less useful after damage: they are as resilient in relative terms and keep much better gaits in absolute terms, so that the quality of the repertoire, more than the adaptation algorithm, decides how well a damaged robot walks. ITE finds a good gait in a handful of trials, orders of magnitude fewer than RL re-training, but in this setting most of its benefit comes from a good prior: a simple ranking of re-evaluated gaits is almost as good. Future work: descriptors that make the effect of damage smoother, pre-trained critics for RL fine-tuning, and tests on a physical robot.

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A Reproducibility

```
python -m qd_damage.repertoires -algo {me,pgame,dcr1me} -seed S
python -m qd_damage.oracle -runs results/repertoires/<run>
python -m qd_damage.adaptation -runs results/repertoires/<run>
python -m qd_damage.rl_retrain -scenario <s> -variant {scratch,finetune} -seed S
python -m qd_damage.analysis.report
```

Versions: QDax 0.5.0, JAX 0.4.28, Brax 0.10.4 (pinned; see `requirements-lock.txt`). Repertoires were built on Kaggle (2× Tesla T4). Every run stores its configuration, seed, code commit, library versions and device.